Lecture Notes

Lecture 1

Vector Spaces and Subspaces

You already know how to add vectors in $\mathbb{R}^3$, multiply matrices, and solve linear systems. This course will carry your knowledge to a more abstract level. The first step is the most important one. Instead of asking *what* a vector is—an arrow? a column of numbers?—we will declare that a vector is **anything that obeys the rules of vector addition and scalar multiplication.** Nothing else about a vector's nature will matter.

This sounds like we are throwing away information, but it is exactly the opposite. By proving theorems from the rules alone, every result we obtain applies *simultaneously* to arrows in space, to matrices, to polynomials, to functions, and to the amplitude and state-vector spaces used in quantum mechanics.

Let us briefly survey where the course is going. Quantum-mechanical state spaces are complex vector spaces, so linear combinations of state vectors express superposition. A normalized vector represents a physical state only up to phase. Mixed states need density operators. Each of these terms is defined when it is first needed: inner products and normalization in Lecture 7, phase and pure states in Lecture 8, density operators in Lecture 10. For now, read them only as pointers to what lies ahead.

The aim of this lecture is to make the linear structure precise: to define fields, vector spaces and subspaces, and to learn how to build new spaces out of old ones by taking sums and direct sums.

Remark 1.1 (Standing scope for Lectures 1–6). Unless a theorem or displayed example explicitly says otherwise, we work with finite-dimensional vector spaces over $\mathbb{R}$ or $\mathbb{C}$. Infinite-dimensional examples — spaces of functions, polynomials or sequences — are only illustrations. They do not by themselves extend formulas proved for finite dimension.

Learning objectives

By the end of this lecture you will be able to:

- State the vector-space axioms and check (or refute) them for concrete sets—tuples, matrices, polynomials, functions.
- Explain why the field of numbers ($\mathbb{R}$ or $\mathbb{C}$) is part of the data of a vector space.
- Apply the three-part subspace test to decide whether a subset is a subspace.
- Form sums and direct sums, and use the zero-test (or $U \cap W = \{0\}$) to decide directness.

1 Scalars: the field

1.1 Fields

Before we can *scale* a vector we must say what numbers we are allowed to scale by. Those numbers form a *field*.

Definition 1.2 (Field). A *field* $\mathbb{F}$ is a set with two operations, addition and multiplication, that are both commutative and associative and are linked by the distributive law $a(b + c) = ab+ac$. Each operation has an identity element (0 and 1, with $1 \neq 0$), every element has an *additive inverse*, and every *nonzero* element has a *multiplicative inverse*—so we may divide by any nonzero number.

The last property is the most important one: in a field you can divide by any nonzero number. Nearly every computation in linear algebra rests on it. In this course we will use only two fields,

$$\mathbb{R} \quad \text{(the real numbers)} \qquad \text{and} \qquad \mathbb{C} \quad \text{(the complex numbers)},$$

and when a statement holds for both we write $\mathbb{F}$ and call its elements numbers or *scalars*.

Remark 1.3. The integers $\mathbb{Z}$ are *not* a field: you cannot divide 1 by 2 and stay in $\mathbb{Z}$. So although $\mathbb{Z}$ has perfectly good addition and multiplication, "a vector space over $\mathbb{Z}$" is not something we allow—we would lose the ability to rescale freely.

1.2 Complex numbers, briefly

Because the standard quantum formalism uses complex amplitudes, we pause to recall the arithmetic we will lean on all semester. A complex number is written $z = a+bi$ with $a, b \in \mathbb{R}$ and $i^2 = -1$; its *real* and *imaginary* parts are a and b, its *conjugate* is $\bar{z} = a - bi$, and its *modulus* is

$$|z| = \sqrt{z\bar{z}} = \sqrt{a^2 + b^2}.$$

Division is carried out by "rationalising" with the conjugate, e.g.

$$\frac{1}{2+i} = \frac{2-i}{(2+i)(2-i)} = \frac{2-i}{5}.$$

Every nonzero complex number also has a *polar form*

$$z = r\,e^{i\theta} = r(\cos\theta + i\sin\theta), \qquad r = |z| > 0, \tag{1.1}$$

where the argument θ is determined modulo 2π. The number 0 has no argument and is handled separately. For nonzero factors multiplication is then strikingly simple: moduli multiply and arguments add modulo 2π, $r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2\, e^{i(\theta_1+\theta_2)}$.

Example 1.4 (Polar form in action). Write $1+i$ in polar form and compute $(1+i)^4$. Here $r = \sqrt{1^2+1^2} = \sqrt{2}$ and $\theta = \frac{\pi}{4}$, so $1+i = \sqrt{2}\,e^{i\pi/4}$. Raising to the fourth power multiplies the argument by 4 and the modulus to the fourth:

$$(1+i)^4 = (\sqrt{2})^4 e^{4\cdot(i\pi/4)} = 4\,(\cos\pi + i\sin\pi) = -4.$$

The polar form turned a messy binomial expansion into one line. ◁

Remark 1.5. The factor $e^{i\theta}$ is a complex number of modulus 1. In quantum mechanics such a factor is called a *phase*. Interference—the phenomenon behind the double-slit experiment—is a consequence of the product of different phases, which is why the standard formulation of quantum mechanics uses complex spaces. Real numbers alone are not enough to describe quantum phenomena.

2 Vector spaces

Definition 1.6 (Vector space). A *vector space over* $\mathbb{F}$ is a set V together with an *addition* $(u, v) \mapsto u+v \in V$ and a *scalar multiplication* $(a, v) \mapsto av \in V$ (for $a \in \mathbb{F}$ and $u, v \in V$) satisfying, for all $u, v, w \in V$ and $a, b \in \mathbb{F}$:

1. $u + v = v + u$ (commutativity);
2. $(u+v)+w = u+(v+w)$ and $(ab)v = a(bv)$ (associativity);
3. there exists $0 \in V$ with $v + 0 = v$ (additive identity);

4. for each v there is $w \in V$ with $v + w = 0$ (additive inverse);
5. $1\,v = v$ (multiplicative identity);
6. $a(u + v) = au + av$ and $(a + b)v = av + bv$ (distributivity).

Elements of V are *vectors*. A vector space over $\mathbb{R}$ is a *real* vector space; over $\mathbb{C}$, a *complex* vector space. We abbreviate $u + (-v)$ as $u - v$.

Remark 1.7. Notice what the definition does *not* say: it never specifies what a vector is. A vector might be a tuple of numbers, a matrix, a polynomial, a function, or a quantum amplitude/state vector. All that is required is that the axioms hold. Prove a theorem from the rules and you have proved it everywhere at once—that is the whole strategy of the subject.

The axioms resemble the ordinary rules of arithmetic, so several "obvious" facts must now be *derived* rather than assumed. The proofs are short but worth following closely: they are your first taste of reasoning from axioms.

Proposition 1.8 (Elementary consequences). Let V be a vector space over $\mathbb{F}$. Then

1. the additive identity 0 is unique, and each $v \in V$ has a unique additive inverse, written $-v$;
2. $0\,v = 0$ for every $v \in V$ (the left 0 is the scalar, the right 0 the zero vector);
3. $a\,0 = 0$ for every $a \in \mathbb{F}$;
4. $(-1)v = -v$ for every $v \in V$.

Proof. (1) If 0 and $0'$ are both additive identities then $0' = 0' + 0 = 0 + 0' = 0$. Here we used the axiom on the existence of an additive identity, and commutativity. For inverses, if $v + w = 0$ and $v + w' = 0$ then

$$w = w + 0 = w + (v + w') = (w + v) + w' = 0 + w' = w'.$$

(2) By distributivity, $0\,v = (0 + 0)v = 0\,v + 0\,v$. Adding the inverse of $0\,v$ to both sides gives $0 = 0\,v$.

(3) Similarly $a\,0 = a(0 + 0) = a\,0 + a\,0$, so $a\,0 = 0$.

(4) We show $(-1)v$ is *an* additive inverse of v; uniqueness from (1) then identifies it with $-v$. Indeed, using the multiplicative-identity axiom, distributivity and (2),

$$v + (-1)v = 1\,v + (-1)v = \big(1 + (-1)\big)v = 0\,v = 0. \qquad \blacksquare$$

Using addition and scalar multiplication we can now define what is arguably the most important construction in all of linear algebra.

Definition 1.9 (Linear combination). A *linear combination* of vectors $v_1, \ldots, v_m \in V$ is any vector of the form $a_1 v_1 + \cdots + a_m v_m$ with scalars $a_1, \ldots, a_m \in \mathbb{F}$.

Almost everything ahead is phrased in this language. In physics terms, the superposition principle says that linear combinations live in the ambient amplitude/state-vector space. A nonzero combination must still be normalized to represent a pure state, and vectors differing by an overall phase represent the same pure state. The set of vectors obtainable as linear combinations of given vectors is called their *span*. That will be the subject of the next lecture.

3 A gallery of examples

The abstract definition pays off by covering a great variety of objects. In every example below, addition and scalar multiplication are performed *entrywise* (for functions, pointwise), so the axioms hold automatically: they already hold for the scalars, and there is almost nothing left to check.

Example 1.10 (Coordinate space $\mathbb{F}^n$). $\mathbb{F}^n$ is the set of n-tuples $(x_1, \dots, x_n)$ with $x_k \in \mathbb{F}$, added and scaled componentwise. The zero vector is $(0, \dots, 0)$. Taking $\mathbb{F} = \mathbb{R}$ and $n = 3$ recovers the familiar $\mathbb{R}^3$; taking $\mathbb{F} = \mathbb{C}$ gives the amplitude/state-vector space for a finite n-level quantum model (for $n = 2$, a qubit). Normalized vectors represent pure-state rays rather than distinct physical states. ◁

Example 1.11 (Matrices and polynomials). $M_{m\times n}(\mathbb{F})$, the $m \times n$ matrices over $\mathbb{F}$, is a vector space under entrywise operations, with the zero matrix as zero vector. So is $\mathcal{P}(\mathbb{F})$, the set of all polynomials $p(t) = a_0 + a_1 t + \cdots + a_d t^d$ with coefficients in $\mathbb{F}$, and so is its subset $\mathcal{P}_m(\mathbb{F})$ of polynomials of degree *at most* m. The words "at most" matter: the polynomials of degree *exactly* m are *not* a vector space, because $t^m + (-t^m)$ has smaller degree. ◁

Example 1.12 (Functions and sequences). Let S be a nonempty set. Denote by $\mathbb{F}^S$ the set of all functions $f\colon S \to \mathbb{F}$, with the operations $(f + g)(x) = f(x) + g(x)$ and $(af)(x) = a\,f(x)$. This is a vector space whose zero vector is the function that is 0 everywhere (the zero function). For instance, in $\mathbb{R}^{\mathbb{R}}$ with $f = \sin$ and $g = \cos$, the vector $2f - 3g$ is the function $x \mapsto 2\sin x - 3\cos x$, taking the value -3 at $x = 0$. Two special cases recur constantly:

- $S = \{1, 2, 3, \dots\}$ gives the space $\mathbb{F}^\infty$ of infinite *sequences* $(x_1, x_2, \dots)$;
- $S = [a, b]$ gives function spaces on an interval, among them the space $\mathscr{C}[a, b]$ of continuous functions. Quantum-mechanical wavefunctions are complex square-integrable functions (they need not be continuous); we will meet their space L^2 in Lecture 13. ◁

Example 1.13 (The same set, two different spaces). The set $\mathbb{C}$ can be regarded as a *complex* vector space (scalars from $\mathbb{C}$) *or* as a *real* vector space (scalars from $\mathbb{R}$ only). These are genuinely different structures: over $\mathbb{C}$ every element is obtained by scaling the vector 1, whereas over $\mathbb{R}$ we need both of the vectors 1 and i to reach every element. More generally $\mathbb{C}^n$ may be viewed as a real vector space, and doing so doubles the number of independent directions. The field the scalars are taken from is part of the definition of a vector space. This remark will become important as soon as we count dimensions, in the next lecture. ◁

Example 1.14 (State spaces in physics). Two examples tie directly to what you know and to what is coming.

1. *Homogeneous linear systems.* The solution set $\{x \in \mathbb{F}^n : Ax = 0\}$ is a vector space: if $Ax = 0$ and $Ay = 0$ then $A(ax + by) = aAx + bAy = 0$. You already met this set when solving $Ax = 0$ by Gaussian elimination: it is the null space of A, $\operatorname{null} A$.
2. *The harmonic oscillator.* Choose a real number $\omega > 0$. The twice-differentiable solutions $y\colon \mathbb{R} \to \mathbb{R}$ of $\ddot{y} + \omega^2 y = 0$ form a real vector space: if y_1, y_2 are solutions then so is $ay_1 + by_2$ (differentiation is a linear operation). The solution space is $\{A\cos\omega t + B\sin\omega t : A, B \in \mathbb{R}\}$; we will see next lecture that it is two-dimensional. Closure under linear combinations is the *superposition principle.* ◁

Example 1.15 (A counterexample). Not every set is a vector space. The line (Figure 1)

$$L = \{(x, y) \in \mathbb{R}^2 : y = x + 1\}$$

is not a vector space: it misses the origin, so it has no zero vector. Nor is it closed under addition: adding the points $(0, 1)$ and $(1, 2)$ of L gives $(1, 3)$, which is off the line. In the next section we will see that the requirement "passes through the origin" is exactly the repair. ◁

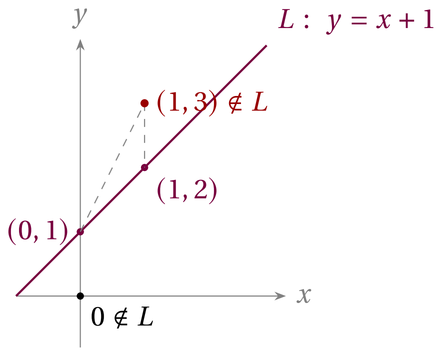

Figure 1: A line not through the origin fails both the zero-vector and closure conditions. The sum of two points on L leaves L.

4 Subspaces

Many vector spaces arise sitting *inside* a larger one: a line through the origin inside $\mathbb{R}^3$, the continuous functions inside all functions.

Definition 1.16 (Subspace). A subset $U \subseteq V$ is a *subspace* of V if U is itself a vector space under the addition and scalar multiplication inherited from V.

There is no need to check all the axioms each time. Almost every axiom is inherited from V automatically. Only three conditions can fail. We isolate them in a separate theorem.

Theorem 1.17 (Subspace test). A subset $U \subseteq V$ is a subspace of V if and only if

1. $0 \in U$; (the zero-vector condition);
2. $u + w \in U$ whenever $u, w \in U$ (closure under addition);
3. $au \in U$ whenever $a \in \mathbb{F}$ and $u \in U$ (closure under scaling).

Proof. If U is a subspace it is a vector space, hence closed under both operations and in possession of a zero (which must coincide with the 0 of V). Conversely, assume (1)–(3). Conditions (2) and (3) guarantee the two operations map U into U, and (1) supplies the additive identity. Each $u \in U$ has its inverse inside U as well, since $-u = (-1)u \in U$ by (3) and Proposition 1.8(4). The remaining axioms (commutativity, associativity, distributivity, $1u = u$) are identities that already hold for all of V, so in particular they hold on U. Hence U is a vector space, i.e. a subspace. ■

Remark 1.18. Condition (1) may be replaced by "U is nonempty": pick any $u \in U$, then $0\,u = 0 \in U$ by closure. In practice the quickest disqualifier runs the other way—if $0 \notin U$, stop, it is not a subspace.

Example 1.19 (Each test condition is necessary). None of the three conditions is redundant. We give three subsets of $\mathbb{R}^2$. Each of them fails exactly one condition.

- The integer lattice $\mathbb{Z}^2 = \{(m, n) : m, n \in \mathbb{Z}\}$ contains 0 and is closed under addition, but not under scaling, since $\frac{1}{2}(1, 0) = (\frac{1}{2}, 0) \notin \mathbb{Z}^2$.
- The union of the first and third quadrants of the plane $\{(x, y) : xy \geq 0\}$ contains 0 (the point $(0, 0)$) and is closed under scaling, but not under addition: $(1, 0)$ and $(0, -1)$ both lie in it, yet their sum $(1, -1)$ does not, since for that point $xy = -1 < 0$.
- The empty set $\varnothing$ is vacuously closed under addition and scaling (there are no vectors to test), but it does not contain 0.

So any subspace must satisfy all three conditions. ◁

Example 1.20 (Subspaces and non-subspaces). Inside $\mathbb{R}^3$, every line and plane *through the origin* is a subspace; a line not through the origin is not. Inside $M_{n\times n}(\mathbb{F})$, each of the following is a subspace, because its defining condition survives addition and scaling: the symmetric matrices ($A^\mathsf{T} = A$), the antisymmetric matrices ($A^\mathsf{T} = -A$), the upper-triangular matrices, and the trace-zero matrices ($\operatorname{tr} A = 0$). By contrast for $n \geq 2$ the singular matrices $\{A : \det A = 0\}$ are *not* a subspace: $\operatorname{diag}(1, 0, \ldots, 0)$ and $\operatorname{diag}(0, 1, \ldots, 1)$ are singular, yet their sum is I_n, and $\det I_n = 1$. For $n = 1$ the singular set is just $\{0\}$, which is a subspace. ◁

Example 1.21 (A subspace defined by a linear condition). Is $U = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$ a subspace of $\mathbb{F}^3$? Run the test. First $0 = (0, 0, 0)$ satisfies $0 = 5 \cdot 0$, so $0 \in U$. Next, if $x_1 = 5x_3$ and $y_1 = 5y_3$ then $(x_1 + y_1) = 5(x_3 + y_3)$, so $x + y \in U$; and $(ax_1) = 5(ax_3)$, so $ax \in U$. All three conditions hold, so U is a subspace. What made it work is that the defining equation is *linear and homogeneous*—contrast the non-example $y = x + 1$ of Example 1.15, whose constant term destroyed closure. ◁

5 Sums and direct sums

Given two subspaces $U, W \subseteq V$, how do we combine them into a third? There are two natural attempts—intersection and union—and they behave very differently.

> **Proposition 1.22 (Intersection is a subspace).** If U and W are subspaces of V, then $U \cap W$ is a subspace of V.

Proof. Both U and W contain 0, so $0 \in U \cap W$. If $x, y \in U \cap W$ and $a \in \mathbb{F}$, then $x + y$ and ax lie in U (because $x, y \in U$ and U is a subspace) and likewise in W, hence in $U \cap W$. The three conditions of the subspace test (Theorem 1.17) hold, so $U \cap W$ is a subspace. ■

Example 1.23 (Two planes meet in a line). In $\mathbb{R}^3$ take the planes

$$U = \{(x, y, z) : z = 0\} \quad \text{and} \quad W = \{(x, y, z) : y = 0\}.$$

Both planes are subspaces. Their intersection $U \cap W = \{(x, 0, 0)\}$ is the x-axis — again a subspace, exactly as Proposition 1.22 promises. ◁

The *union* of subspaces is almost never a subspace: the union of the two coordinate axes in $\mathbb{R}^2$ contains $(1, 0)$ and $(0, 1)$ but not their sum $(1, 1)$ (Figure 2). We will therefore build a new subspace from U and W not by taking their union, but by taking their *sum*.

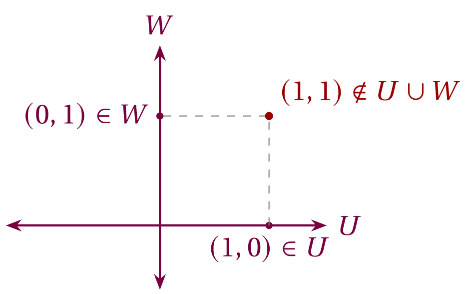

Figure 2: The union of two subspaces need not be a subspace: the sum of a nonzero point on the x-axis and a nonzero point on the y-axis lies on neither axis.

Definition 1.24 (Sum of subspaces). If $U_1, \ldots, U_m$ are subspaces of V, their *sum* is

$$U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_k \in U_k\}.$$

A quick check with the subspace test shows $U_1 + \cdots + U_m$ is again a subspace; in fact it is the *smallest* subspace of V that contains every U_k (any subspace containing all the U_k must contain all their sums).

A sum is most useful when each vector decomposes in only *one* way.

Definition 1.25 (Direct sum). The sum $U_1 + \cdots + U_m$ is a *direct sum*, written $U_1 \oplus \cdots \oplus U_m$, if every vector in it has *exactly one* expression as $u_1 + \cdots + u_m$ with $u_k \in U_k$.

To test directness we need not inspect every vector—only the zero vector.

Proposition 1.26 (Zero-test for directness). $U_1 + \cdots + U_m$ is direct if and only if the only way to write $0 = u_1 + \cdots + u_m$ with $u_k \in U_k$ is $u_1 = \cdots = u_m = 0$.

Proof. If the sum is direct, 0 has a unique decomposition; the all-zero one works, so it is the only one. Conversely, suppose 0 decomposes only trivially and take any v with two decompositions $v = \sum_k u_k = \sum_k u'_k$. Subtracting, $0 = \sum_k (u_k - u'_k)$ with $u_k - u'_k \in U_k$, so by hypothesis each $u_k - u'_k = 0$; the two decompositions coincide. ∎

For two subspaces the criterion takes a memorable geometric form.

Corollary 1.27 (Two-subspace criterion). For subspaces $U, W \subseteq V$,

$$U + W \text{ is a direct sum} \iff U \cap W = \{0\}.$$

Proof. ($\Rightarrow$) If $v \in U \cap W$ then $0 = v + (-v)$ with $v \in U$ and $-v \in W$; directness forces $v = 0$. ($\Leftarrow$) If $U \cap W = \{0\}$ and $0 = u + w$ with $u \in U$, $w \in W$, then $u = -w \in U \cap W$, so $u = 0$ and hence $w = 0$; Proposition 1.26 then gives directness. ∎

Example 1.28 (Decomposing $\mathbb{R}^3$). Let $U = \{(x, y, 0)\}$ be the xy-plane and $W = \{(0, 0, z)\}$ the z-axis. Every vector splits as $(x, y, z) = (x, y, 0) + (0, 0, z)$ (Figure 3), and $U \cap W = \{0\}$, so $\mathbb{R}^3 = U \oplus W$. The splitting is unique, so the sum is direct. ◁

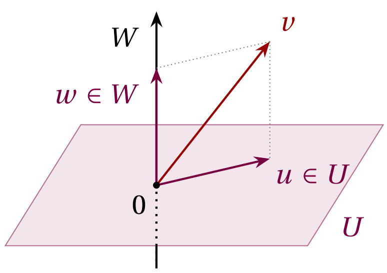

Figure 3: $\mathbb{R}^3 = U \oplus W$: each v splits uniquely into a vector u in the plane U and a vector w on the axis W.

Example 1.29 (Symmetric and antisymmetric matrices). Let Sym and Skew be the symmetric and antisymmetric subspaces of $M_{n\times n}(\mathbb{F})$. Any matrix splits as

$$A = \underbrace{\tfrac{1}{2}(A + A^{\mathsf{T}})}_{\in \mathrm{Sym}} + \underbrace{\tfrac{1}{2}(A - A^{\mathsf{T}})}_{\in \mathrm{Skew}}, \tag{1.2}$$

so $M_{n\times n}(\mathbb{F}) = \mathrm{Sym} + \mathrm{Skew}$; and a matrix that is both symmetric and antisymmetric satisfies $A = A^{\mathsf{T}} = -A$, forcing $A = 0$, so the intersection is trivial. By Corollary 1.27, $M_{n\times n}(\mathbb{F}) = \mathrm{Sym} \oplus \mathrm{Skew}$. For example,

$$\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

is the unique symmetric-plus-antisymmetric decomposition prescribed by (1.2). This very splitting underlies the decomposition of a physical strain or field gradient into a deformation and a rotation. ◁

Remark 1.30 (Why this matters in physics). In quantum mechanics a measured quantity has several possible values. Each of them has its own subspace of states. Those subspaces form a direct sum, so any state can be split in exactly one way into components — one for each possible value of the measurement. How "large" a component is gives the probability of obtaining the corresponding value. How to measure the size of a component we will learn later, when we define the inner product and the norm; the splitting itself is precisely today's direct sum. We will return to all of this in Lecture 10.

Summary of main concepts

A vector space is a set closed under addition and scalar multiplication and satisfying the list of axioms. The elements of that set are called vectors. What those vectors "are" is irrelevant. Coordinate n-tuples, matrices, polynomials, functions and quantum state-vector spaces are all examples of vector spaces. A subspace is a subset of a vector space that contains 0 and is closed under both operations; those are exactly the three things to check. Subspaces combine through sums, and a sum is *direct* precisely when the zero vector has only one (all-zero) decomposition. For two subspaces this means that their intersection is $\{0\}$.

For three or more subspaces it is not enough to check the pairwise intersections. In $\mathbb{R}^2$ take three lines through the origin: the x-axis, the y-axis and the line $y = x$. Any two of them meet only at 0, yet $(1, 0) + (0, 1) - (1, 1) = (0, 0)$ — here we have decomposed the zero vector nontrivially. Quantum mechanics uses $\mathbb{C}$ because complex amplitudes carry a phase; real numbers alone are not enough to describe that multiplication of phases.

- **Field**—scalars you can add, multiply, and divide by; for us always $\mathbb{R}$ or $\mathbb{C}$.
- **Vector space**—a set closed under addition and scalar multiplication satisfying the axioms—the nature of its elements, the vectors, is irrelevant.
- **Linear combination**—$a_1v_1 + \cdots + a_mv_m$; quantum superposition occurs in the ambient amplitude/state-vector space.
- **Subspace**—a subset containing 0 and closed under both operations (the three-part test).
- **Sum and direct sum**—$U_1 + \cdots + U_m$; *direct* (written $\oplus$) when every vector decomposes uniquely — it is enough to check the zero vector.

The problems for this lecture are in the sheet `exercises-1.pdf`.